Anirudh Bhat

ep24btech11004@iith.ac.in | anirudh.b.0616@gmail.com | +91 8310650531 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.58
Bangalore International Academy (BIGI)	Class XII (CBSE)	2024	93.80%
National Public School, Banashankari	Class X (CBSE)	2022	93.60%

OPEN SOURCE PROJECTS

Docright.nvim - AI Contextual Code Assistant (Neovim Plugin) (GitHub)

| Lua, Neovim API, Ollama, LLM Integration, Code Analysis

- Neovim plugin that seamlessly integrates **local LLM** models (via Ollama) directly into your development workflow, enabling **real-time code interaction** and contextual queries without leaving the editor.
- Implemented **intelligent code context extraction** and prompt engineering to feed relevant code snippets and surrounding logic to LLM models, ensuring contextually accurate and relevant responses.
- Designed **dual-mode interface** with document generation, one for creating code explanations and ask mode for interactive code-related questioning, so you can go down rabbit holes!
- Built modular architecture **supporting multiple LLM providers** and configurable model endpoints, enabling developers to deploy with different local models (Qwen, Llama, etc.) based on performance and hardware constraints.

PROJECTS

Gamma-Ray Burst Temporal Analysis (Research Project) (Report)

| Python, NumPy, SciPy, Astropy, Nonlinear Dynamics, Statistical Analysis

- Analysed Gamma-Ray Burst datasets using pulse-detection and **time-series analysis** techniques to study burst temporal evolution.
- Implemented phase-space reconstruction and **Lyapunov exponent estimation** to investigate evidence of chaotic dynamics.
- Developed data-processing workflows for **noise reduction**, **peak detection and waiting-time analysis** of burst activity.
- Compared dynamical behaviour across GRB observations using quantitative and visual analyses to assess chaos indicators.

Modal Text Editor - Chim (GitHub)

| C, POSIX APIs, ANSI Terminal Control, Make, Git, Homebrew

- Lightweight modal text editor in C with **Vim-inspired** motions and **keyboard-driven** binds with **low-latency** terminal interaction.
- Dynamic row-based text storage and viewport management systems supporting **multi-line editing**, and **file persistence**.
- Built **incremental search** and **syntax highlighting** systems using lightweight lexical parsing and real-time match rendering.
- Designed a **custom** terminal rendering pipeline with line numbering, status bars, and **optimised screen redraw behaviour**.
- Packaged and distributed the editor through a custom Homebrew tap for **streamlined versioned releases** on macOS.

Pulsar Population Evolution Analysis (Research Project)

| Python, NumPy, Pandas, Astropy, Statistical Modelling, Astrophysics

- Analysed ATNF pulsar data to study **pulsar population evolution** in the period-period derivative parameter space.
- Implemented population-statistics methods investigating birthrates and **evolutionary trends** across spin-down models.
- Performed **data processing, visualisation**, and **analysis** comparing theoretical evolution models with observations..

- Developed tools for astronomical catalogue analysis, model evaluation, and population inference.

Gravity Wave Identification using Bayesian MCMC (GitHub)

| Python, NumPy, SciPy, Bayesian inference, Numerical Methods

- Investigated **Bayesian techniques** for gravitational-wave-like signal recovery under significant observational noise.
- Applied **Metropolis-Hastings MCMC sampling** to estimate physical model parameters and characterise posterior uncertainties.
- Analysed parameter degeneracies and correlations through **covariance studies** and **posterior distribution modelling**.
- Validated inference results using synthetic datasets and reconstructed signal comparisons.

POSITIONS OF RESPONSIBILITY

Assistant Manager - Ideation Domain, E-Cell (Entrepreneurship Club)

- Assisted in **speaker outreach** and communication for E-Summit, engaging with industry leaders and entrepreneurs.
- Worked closely with other teams to identify and **solve operational challenges** during event planning and execution.

Core member - Writing Club, The Literary Society of IITH

- Represented the college in inter-collegiate literary events, securing **multiple awards** and cash prizes across various institutions
- Organised and hosted writing and literary events for the student community
- Facilitated judging processes, participant evaluation and assessments for club events
- **Currently the Writing Club Head**

SKILLS

Programming Languages: Python, C, C++, Lua, HTML, SQL, Typst, LaTeX

Tools: Git, Vim, Bash(Linux Shell), Unix, Tmux, Homebrew, Make

Libraries: NumPy, Astropy, Pandas, Matplotlib, SciPy, Nolds

RELEVANT COURSES

Computer Science : Data Structures and applications, Introduction to Programming(C), Artificial Intelligence.

Mathematics : Advanced Mathematical Physics, Elementary Linear Algebra, Differential Equations, Calculus I and II, Intro to Statistics/Probability, Writing and Presenting Math

Physics : Special Relativity, Computational Physics, Advanced Mathematical Physics, Modern Physics, Quantum Mechanics,